

IMO Training

Combinatorics

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Topic 1 Counting

1.1 The four arithmetic operations

We frequently use the four basic arithmetic operations in counting. Can you give examples?

- Addition rule and multiplication rule — the two very basic operations
- Subtraction — counting the opposite
- Division — to correct for overcounting

Another reason for subtraction is to correct for overcounting, summarised by the **inclusion-exclusion principle**.

1.2 Permutations of n objects

How many **permutations** of n objects are there if

- the objects are distinct?
- some of the objects are the same?
- we only need to permute *some* (rather than *all*) of the objects?
- the permutation is *circular*?

1.3 The ' n choose r ' problem

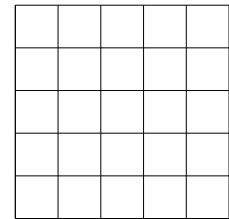
How many ways are there to choose r objects out of n (distinct) objects?

	Order matters	Order does not matter
Repetition allowed	(a)	(b)
Repetition not allowed	(c)	(d)

1.4 Miscellaneous examples

1. (PCIMC 2003) How many 4-digit positive integers have the product of their digits even?

2. The figure shows a 5×5 square grid.



- (a) How many squares can be found?
- (b) How many rectangles (including squares) can be found?
- (c) What is the sum of the areas of the rectangles in (b)?
- (d) In how many ways can one travel from the lower left hand corner to the upper right hand corner, if in each step one can only go upward or rightward?

3. There are some red, yellow, green and blue marbles. There are 20 marbles of each colour.

- (a) If we need to choose two marbles of different colours, how many choices are there?
- (b) If we need to choose ten marbles, how many different combinations of colours are there?
- (c) What will be the answer to (b) if we require at least two red marbles?

4. How many ways are there to divide a class of 40 students into eight groups of 5?
5. In how many ways can we choose 13 cards from a pack of 52 so that there is at least one void?
6. In the Mark Six lottery in Hong Kong, what is the probability of winning the various prizes?
7. When 5 cards are drawn from a pack of 52, what is the probability of getting the various combinations?

Exercise 1

1. How many ways are there to seat 8 boys and 4 girls at a round table if no two girls may sit next to each other?
2. Six men and six women are to be paired up. Two men and two women insist pairing up with someone of the same gender. Another two men and two women insist pairing up with someone of the opposite gender. How many ways of pairing are there?
3. (IMO HK Prelim 1999) A solid pyramid $VABCD$, with a quadrilateral base $ABCD$, is to be coloured on each of the five faces such that no two faces with a common edge will have the same colour. If five different colours are available, what is the number of ways to colour the pyramid?
4. (AIME 2006) Let $(a_1, a_2, a_3, \dots, a_{12})$ be a permutation of $(1, 2, 3, \dots, 12)$ for which

$$a_1 > a_2 > a_3 > a_4 > a_5 > a_6 \quad \text{and} \quad a_6 < a_7 < a_8 < a_9 < a_{10} < a_{11} < a_{12}.$$

An example of such a permutation is $(6, 5, 4, 3, 2, 1, 7, 8, 9, 10, 11, 12)$. Find the number of such permutations.

5. (AIME 2008, modified) There are two distinguishable flagpoles, and there are 19 flags, of which 10 are identical blue flags, and 9 are identical green flags. In how many different ways can we put all the flags onto the flagpoles in which each flagpole has at least one flag and no two green flags on either pole are adjacent?
6. (PCIMC 2008) Find the number of 25-digit positive integers satisfying the following two conditions:
 - The digits of the integer consist of exactly 20 copies of '1' and 5 copies of '2'.
 - If we remove the last k digits from the integer (where k is any positive integer less than 25), the resulting integer has at least as many '1's as '2's.
7. (CMO 2001) On a circle with circumference 24 there are 24 equally spaced points, marked 1 to 24. In how many ways can we choose 8 out of these 24 points so that the length of the minor arc joining any two of the chosen points is not equal to 3 or 8?

8. (CGMO 2008) Given a 2008×2008 chessboard, each cell of which has a different colour. Now one of the 4 letters C, G, M, O is to be filled into each cell. If every 2×2 grid on the chessboard contains all 4 letters C, G, M, O, we say that the chessboard is 'harmonic'. How many different 'harmonic' chessboards are there?

Topic 2 Existence and Optimisation

2.1 The pigeonhole principle

The basic form of the **pigeonhole principle** says

- If there are $n + 1$ pigeons and n holes, then _____.

What are some alternative forms?

2.2 Existence and sharpness of bound

Using the pigeonhole principle, we can show that any 51 numbers chosen from the set $\{1, 2, \dots, 100\}$ must satisfy certain *existence* properties. Some possible properties are:

- Two of the chosen numbers have sum 101.
- Two of the chosen numbers are relatively prime.
- One of the chosen numbers divides another chosen number.

In each case, determine whether the bound 51 is *sharp*.

2.3 Structure of an optimisation problem

An *optimisation* problem generally refers to one which asks for the maximum/minimum value of a quantity subject to certain conditions. In general, we need to do three things in answering such a problem:

- Find the answer
- Show that the answer you claimed satisfies the required conditions
- Show the the answer you claimed is indeed the maximum/minimum

(IMO 1977) In a finite sequence of real numbers the sum of any seven successive terms is negative, and the sum of any eleven successive terms is positive. Determine the maximum number of terms in the sequence.

(APMO 1992) Find a sequence of maximal length of non-zero integers in which the sum of any seven consecutive terms is positive and that of any eleven consecutive terms is negative.

2.4 Miscellaneous examples

1. In a square of side length 4, is it possible to choose 28 points so that no circle with centre inside the square and radius 1 covers 4 or more points?
2. Given nine straight lines, each of which divides a square into two quadrilaterals of areas $2:3$. Show that three of these lines meet at a point.
3. (IMO 1972) Prove that from a set of ten distinct two-digit numbers (in the decimal system), it is possible to select two disjoint subsets whose members have the same sum.
4. At least how many integers must be chosen to ensure that there must be two chosen integers whose sum or difference is divisible by 1000?

Exercise 2

1. In a bag there are 1000 balls, numbered 1 to 1000. What is the minimum number of balls that must be drawn from the bag to ensure that you must be able to get two balls whose sum of numbers has unit digit 7?
2. Nine different real numbers are chosen. Show that we can find x and y among them such that $0 < \frac{x-y}{1+xy} < \sqrt{2}-1$.
3. (JMO 1991) In every 16-digit number, show that there is a string of one or more consecutive digits such that the product of these digits is a perfect square.
4. (IMO 2002 HK Selection Test) For an irrational number x , let x' be the integer nearest to x . Define $\langle x \rangle = |x - x'|$. Show that for every irrational number y , the minimum of the numbers $\langle y \rangle, \langle 2y \rangle, \dots, \langle 2001y \rangle$ is less than $\frac{1}{2001}$.
5. In a school, there are 601 students, no two of whom have the same height. They are queued up in a row in random order. Show that the teacher can always find at least one of the following:
 - 21 students in the queue whose heights are in ascending order; or
 - 31 students in the queue whose heights are in descending order.
6. (PCIMC 2019) There is a $2 \times n$ table. In each cell a different positive integer is written. If we add up the two numbers in each column, the sums obtained are 2019, 2020, ..., $2018+n$ respectively. Find the greatest possible value of n .
7. (AIME 2010) Let $m \geq 3$ be an integer and let $S = \{3, 4, 5, \dots, m\}$. Find the smallest value of m such that for every partition of S into two subsets, at least one of the subsets contains integers a , b and c (not necessarily distinct) such that $ab = c$.

8. (CHKMO 1999, rephrased) In a class each student studies the same n subjects. One day there is a test in each subject. It is known that for each subject exactly three students fail the test, and for any two distinct subjects exactly one student fail both tests. Determine the smallest possible value of n for which the above conditions imply that exactly one student fails all n tests.
9. (IMO 2012 shortlist) Let $n \geq 1$ be an integer. What is the maximum number of disjoint pairs of elements of the set $\{1, 2, \dots, n\}$ such that the sums of the different pairs are different integers not exceeding n ?

Topic 3 Recurrence Relations

3.1 Examples for motivation

1. (IMO HK Prelim 1991) A flight of stairs consists of 10 steps. A boy can climb one or two steps each time. In how many different ways can the boy climb from the bottom to the top of the stairs?
2. (PCIMC 2006) In a city, all telephone numbers have 8 digits and all telephones have a keyboard in the form as shown. Furthermore, consecutive digits in a telephone number must be adjacent digits on the keyboard (e.g. 85256321). How many different possible telephone numbers are there?

7	8	9
4	5	6
1	2	3

3. (IMO HK Prelim 2001) Find the number of 10-digit positive integers such that
- (a) each digit is either 1 or 2; and
 - (b) there exist two consecutive 1's.
4. (AIME 2008) Consider sequences that consist entirely of A 's and B 's and that have the property that every run of consecutive A 's has even length, and every run of consecutive B 's has odd length. Examples of such sequences are AA , B , and $AABAA$, while $BBAB$ is not such a sequence. How many such sequences have length 14?

3.2 Solving recurrence relations (I) — easy cases

Solving a recurrence relation means we want to find the _____ of the sequence.

Some recurrence relations are easy to solve.

- Write down an arithmetic sequence and a recurrence relation that it satisfies.
- Write down a geometric sequence and a recurrence relation that it satisfies.
- Can you solve the above recurrence relations? What is the role of the **initial conditions**?
- How do the initial conditions for the Fibonacci sequence differ from those for arithmetic and geometric sequences?
- Can you find a second order recurrence relation for a geometric sequence?
- Can you solve the recurrence relation $a_{n+1} = 3a_n - 4$ with $a_1 = 5$?

3.3 Solving recurrence relations (II) — linear homogeneous with constant coefficients

Linear homogeneous recurrence relations with constant coefficients can be solved by the **method of characteristic equations**.

- $a_{n+2} = 4a_{n+1} - 3a_n$ with $a_1 = 1$ and $a_2 = 2$
- $a_{n+2} = 4a_{n+1} - 4a_n$ with $a_1 = 1$ and $a_2 = 3$

3.4 Solving recurrence relations (III) — when a non-homogeneous term is added

In this case, the general solution is the sum of the general solution to the homogeneous part and a particular solution.

- $a_{n+2} = 4a_{n+1} - 4a_n + 2n$ with $a_1 = 1$ and $a_2 = 3$
- $a_{n+2} = 4a_{n+1} - 4a_n + 3^n$ with $a_1 = 1$ and $a_2 = 3$
- $a_{n+2} = 4a_{n+1} - 3a_n + 3^n$ with $a_1 = 1$ and $a_2 = 3$

Exercise 3

- Find the unit digit and the tenth digit (i.e. the two digits immediately next to the decimal point) of the number $(3 + \sqrt{5})^{2345}$.

- (IMO 2019 HK Team Selection Test) Determine all sequences $p_1, p_2, p_3, \dots$ of prime numbers for which there exists an integer k such that the recurrence relation

$$p_{n+2} = p_{n+1} + p_n + k$$

holds for all positive integers n .

- Let a_n be the number of positive integers having digits 1, 3, 4 only and sum of digits equal to n .

(a) Find a_{12} .

(b) Show that a_{2n} is a perfect square for all natural numbers n .

- (IMO 1979) Let A and E be opposite vertices of a regular octagon. A frog starts jumping at vertex A . From any vertex of the octagon except E , it may jump to either of the two adjacent vertices. When it reaches vertex E , the frog stops and stays there. Let a_n be the number of distinct paths of exactly n jumps ending at E . Prove that $a_{2n-1} = 0$ and

$$a_{2n} = \frac{1}{\sqrt{2}} \left[(2 + \sqrt{2})^{n-1} - (2 - \sqrt{2})^{n-1} \right] \text{ for } n = 1, 2, 3, \dots$$

- (CGMO 2008) Let the sequence $x_1, x_2, \dots, x_n, \dots$ of positive real numbers satisfy $(8x_2 - 7x_1)x_1^7 = 8$ and

$$x_{k+1}x_{k-1} - x_k^2 = \frac{x_{k-1}^8 - x_k^8}{(x_k x_{k-1})^7}$$

for $k \geq 2$. Find positive real number a such that, when $x_1 > a$, we have $x_1 > x_2 > \dots > x_n > \dots$, while when $0 < x_1 < a$, the sequence is not monotonic.